Optics & Laser Technology 181 (2025) 111804

Contents lists available at ScienceDirect

Optics and Laser Technology

journal homepage: www.elsevier.com/locate/optlastec

Full length article

Welding between rough copper foil and silica glass using green femtosecond laser

Jingyu Huo [a], Zirong Zeng [a], Jinhui Yuan [a], Minghuo Luo [c], Aiping Luo [a], Jiaming Li [a,*], Huan Yang [b], Nan Zhao [d], Qingmao Zhang [a]

[a] Guangdong Provincial Key Laboratory of Nanophotonic Functional Materials and Devices, South China Normal University, Guangzhou 510006, China
[b] Sino-German College of Intelligent Manufacturing, Shenzhen Technology University, Shenzhen 518118, China
[c] State Key Lab Optoelect Mat & Technol, Sun Yat-sen University, Guangzhou 510275, China
[d] School of Electronics and Information, Guangdong Polytechnic Normal University, Guangzhou 510665, China

ARTICLE INFO

Keywords:
Femtosecond laser
Green laser
Weld
Silica glass
Copper foil

ABSTRACT

The assembly of glass and copper micro devices is widely applied in modern manufacturing industries. Laser welding is an efficient technique. However, weld defects and instability resulting from the low laser absorptivity of copper remain significant challenges. Surface roughness also poses a limitation for optical contact during welding preprocessing. In this work, femtosecond pulse and green light from the second harmonic generation were combined to increase the copper absorptivity. The silica glass and rough copper foil were effectively welded. Under the optimized parameters, a maximum shear strength of 17.19 MPa was obtained. The electron and lattice temperatures during the welding process were simulated using two-temperature model. The microscopical mechanism, element diffusion, and chemical reaction were investigated. A modified region in the glass was formed due to excessive laser energy and scattered subsequent laser pulses. Cu–O-Si bonds were detected on the welds. Welding stability at various temperatures was characterized, with shear strength maintained at approximately 12 MPa after thermal cycling from 0 to 100 °C and heating at 150 °C. This study demonstrated that effective and stable welding of silica glass and rough copper foil can be achieved using green femtosecond lasers.

1. Introduction

High-quality and stable connections between copper and silica glass are among the essential technologies in electronic packaging, aerospace, medical devices, MEMS devices, and semiconductors [1–6]. Traditional connection methods, including adhesive bonding [6], mechanical riveting [7], anodic bonding [8], and solid-state bonding [9], have contributed to the joining of heterogeneous materials. These methods still suffer from inherent limitations such as low efficiency and poor accuracy in industrial processing.

Lasers have proven to be reliable tools for ablating, drilling, and structuring due to their high precision and non-contact nature [10–13]. Laser welding can meet the requirements of modern industrial production and shows great potential in joining metal and glass [14–20]. Utsumi et al. demonstrated that the formation of metal particles is crucial for successful connections in nanosecond laser welding [5]. Zhang et al. investigated the mechanism of welding following micro-arc-oxidation processing of aluminium alloy [21]. In our previous research, optical contact between steel and glass was maintained using an intermediate liquid [22]. Welding of silica glass and stainless steel achieved a maximum strength of 16 MPa under optical contact [16]. A welding method involving the addition of a Cr interlayer between glass and steel was proposed to enhance welding strength [23]. These works demonstrated the possibility of laser welding metal and glass. The addition of an oxide coating and post-process annealing are general processes to reduce stress and ensure welding quality [21,24,25]. Differences in thermal expansion between metal and glass remain a challenge.

Moreover, the optical contact condition (gap < 200 nm) was necessary for laser welding in the aforementioned studies, which is difficult to achieve with industrial-grade copper foil that has a rough surface and high flexibility. In ultrafast laser welding, there is no limitation imposed by the optical contact condition [26], and thermal effects can be more easily controlled. Carter et al. demonstrated the feasibility of welding various glasses and metals using ultrafast laser [27]. Wang

* Corresponding author.
E-mail addresses: jmli@m.scnu.edu.cn (J. Li), yanghuan@sztu.edu.cn (H. Yang).

https://doi.org/10.1016/j.optlastec.2024.111804
Received 26 May 2024; Received in revised form 14 August 2024; Accepted 12 September 2024
Available online 20 September 2024

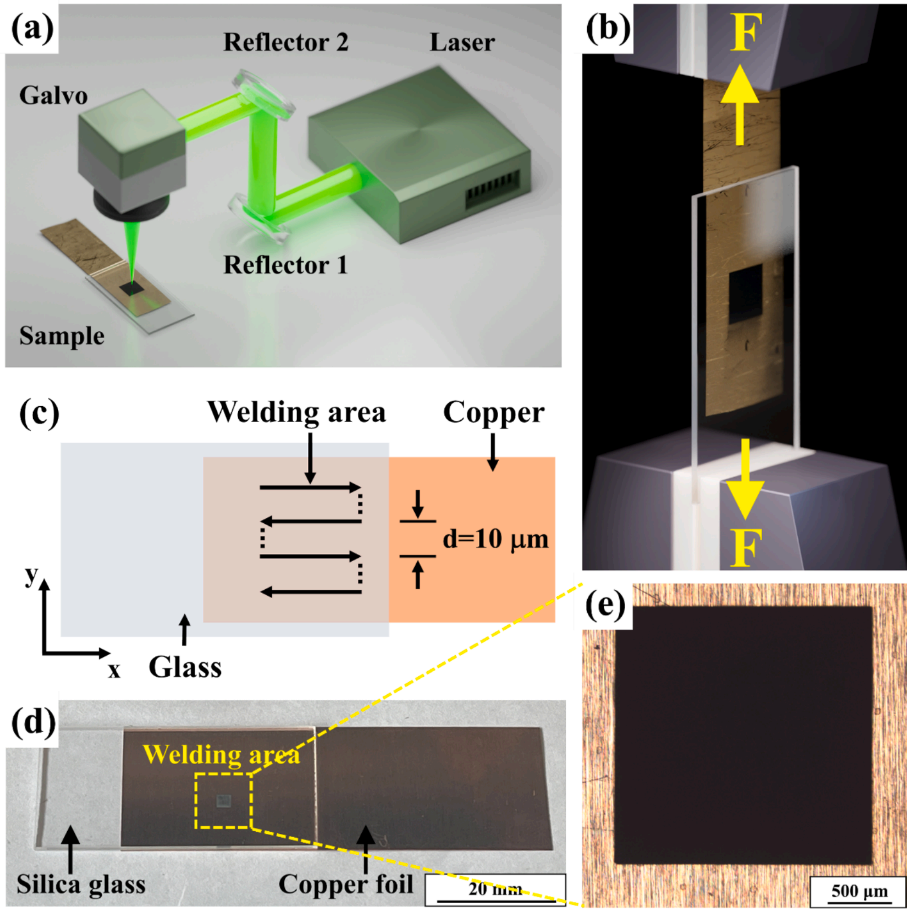

Fig. 1. (a) Diagram of femtosecond laser welding system, (b) test method for shear strength, (c) scanning pattern, (d) photograph of the welded sample, and (e) welding area fixed at 2 mm × 2 mm.

elucidated that the mechanism of welding stainless steel and glass involve wetting, adsorption, and mechanical bonding [28]. Zhang et al. found that the welds were stabilized by a concave-convex structure [29]. Ozeki et al. welded copper and non-alkali glass using infrared fs laser [30]. Zhang et al. demonstrated that metal particles induced by ultrafast laser ablation were the key to successful welding between alumina-silicate glass and copper [31]. Li demonstrated that melting glass filling the gap is a prerequisite for successful glass-to-copper welding [32]. However, the welds defects and instability due to the low absorptivity (<5%) of copper at the fundamental wavelength (800–1080 nm) of ultrafast lasers remain a challenge. Processing quality is significantly improved using green laser radiation, due to its higher absorptivity of copper (~40 %). Copper and other metals have been welded using green lasers to achieve stable joints [33–35]. Microporosity in the copper substrate was achieved using green lasers [36–38]. Moreover, the absorptivity of copper can be further increased through nonlinear effects during interaction with ultrafast lasers. The green femtosecond lasers are likely to provide higher quality welding of copper and silica glass.

This work proposes high-strength and stable welding of rough copper foil and silica glass using a green femtosecond laser (520 nm, 300 fs, 250 kHz). The electron and lattice temperatures during the welding process were simulated using the two-temperature model. The morphology and mechanical properties of the welded samples were investigated. The microscopic mechanisms, element diffusion, and chemical reactions were characterized. The modifications occurring inside the glass during the welding process were observed. The high-temperature resistance of the welded samples was tested. This study aims to facilitate high-quality and stable connections between silica glass and rough copper foil in industrial applications.

2. Experimental procedure

2.1. Materials and facilities

The silica glass (20 mm × 40 mm × 1 mm) and rough copper foil (20 mm × 60 mm × 0.1 mm) were used in this work. The surface roughness of the unpolished copper foil and glass is shown in Tables S1. The compositions of the copper and glass are detailed in Tables S2 and S3, respectively. The thermal physical parameters of the copper and glass are illustrated in Figure S1 and Table S4, respectively.

In this work, a femtosecond laser (Spectra-Physics, Spirit HE 1040–30 SHG) was employed. The wavelength, pulse duration, and repetitive frequency of the laser were 520 nm, 300 fs, and 250 kHz, respectively. The laser beam was focused on the interface between glass and copper foil using an F-Theta lens. The spot diameter was 16 μm. A tensile testing machine (Zwick Roell, z030) was used to measure the shear force to evaluate the welding quality of the samples. The shear strength σ was calculated by Eq. (1):

$$\sigma = \frac{F}{S} \tag{1}$$

where F is shear force; S is welded area. In this work, the welding area S was fixed at 2 mm × 2 mm.

Additionally, a confocal laser scanning microscope (CLSM, OLS5000) was used to measure the surface roughness. Scanning electron microscopy (SEM, ZEISS Gemini 300) was used to observe the morphology of the welds surface. An energy dispersive spectrometry

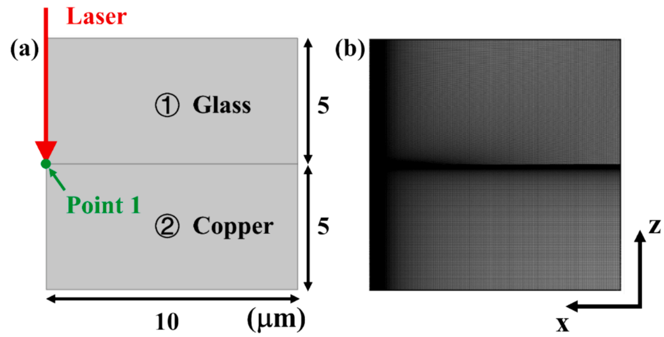

Fig. 2. (a) The geometry model and (b) the mesh.

(EDS, ZEISS Gemini 300) was used to observe the elemental distribution. A spectrophotometer (PerkinElmer, Lambda 1050 +) was used to measure the absorptivity and reflectivity of the copper foil. An X-ray photoelectron spectrometer (XPS, Thermo Fisher Scientific Co. Ltd, Escalab Xi +), equipped with an Al X-ray source (Al Kα, 1486.6 eV), was utilized to characterize the chemical bonds and reactions occurring during the welding process. The testing area was 300 μm × 300 μm. A muffle furnace (Ymaato, FO311C) was used for the high-temperature testing of the welded samples.

Fig. 1 (a) illustrated the laser welding system. A Galvo system was used to control the laser scanning strategy (Fig. 1c). The welded samples were separated by a tensile testing machine to measure the shear force. The clamps were used to secure the samples and prevent deviation during the separation process, as shown in Fig. 1 (b). Each experimental parameter (laser power, scanning speed, focal position) was tested three times. The welded region consisted of horizontal lines spaced 10 μm apart to form a rectangle, which was entirely filled with welds without gaps, as shown in Fig. 1(d) and (e).

Prior to welding, the copper foil and silica glass were ultrasonically cleaned with alcohol for 15 min and then wiped with absorbent paper. The glass was placed on the copper foil surface, maintaining contact under a pressure of 0.16 MPa from the fixture. The laser was focused on the interface between the copper foil and silica glass. Upon reaching the interface, the femtosecond laser was absorbed by both materials. Materials near the focal point were fused to form welds. A connection between the copper foil and silica glass was established.

2.2. Finite element model

A finite element model of femtosecond laser welding of copper foil and silica glass was developed using MATLAB R2021a to characterize the welding process. Due to the symmetry of the laser spot, only half of the welding area was simulated to balance computational time and accuracy, as shown in Fig. 2(a). Area ① represents silica glass, while area ② represents copper. Two probes were positioned at point 1 to record the temperatures of the electrons and lattice, respectively. The larger mesh was positioned away from the glass-copper interface, while the smaller mesh was placed near the interface, as shown in Fig. 2(b). The thermal physical parameters of glass and copper were shown in Table S4 and Figure S1, respectively.

The substrate was heated by a Gaussian heat source [39], described by Eq. (2):

$$Q = \sqrt{\frac{4\ln 2}{\pi}}(1-R)\frac{I_0}{\delta t_p}\exp\left[-\frac{r^2}{r_D^2}-\frac{z}{\delta}-4\ln 2\frac{(t-t_p)^2}{t_p^2}\right] \tag{2}$$

where R is the laser reflectivity on the copper foil; I_0 is laser energy flux; r_D is the spot radius of the laser; t_p is pulse duration; δ is skin depth (10 nm) [40]; $\exp(-\frac{r^2}{r_D^2})$ is the Gaussian spatial distribution function; $\exp(-\frac{z}{\delta})$ is the depth function depending attenuation of the laser pulse

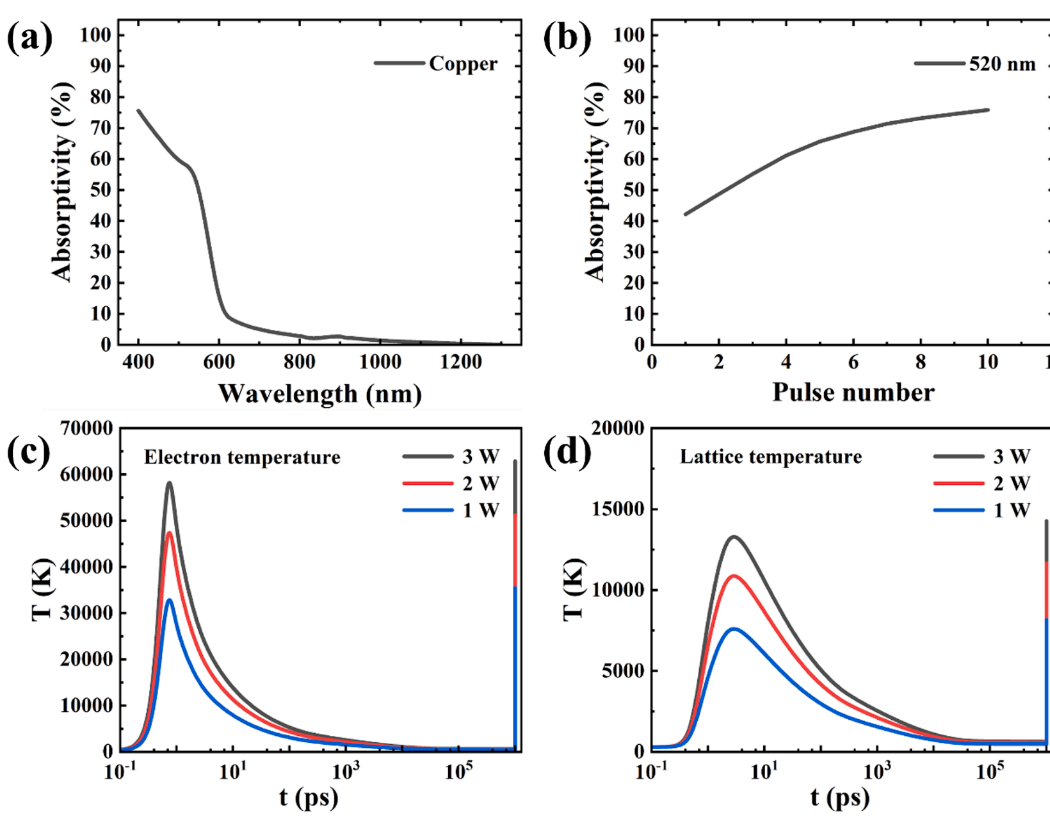

Fig. 3. Absorptivity of the copper foil (a) at different wavelengths, (b) at 520 nm under different laser pulse numbers; electron (c) and lattice (d) temperature at the interface between the silica glass and copper foil.

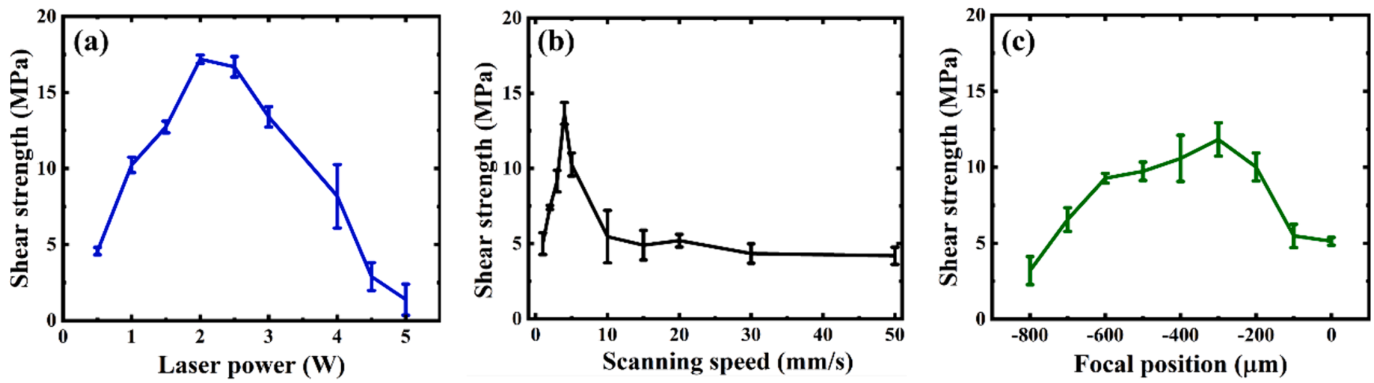

Fig. 4. Effect of different laser process parameters on welding strength: (a) laser power, (b) scanning speed, and (c) focal position.

according to Beer-Lambert Law; $\exp[-4\ln 2\frac{(t-t_p)^2}{t_p^2}]$ is the Gaussian temporal distribution function.

The relaxation processes between the electron and lattice subsystem take place on the scale of sub-picoseconds. The heat transfer process was calculated by two-temperature model (TTM) [41–43], as shown in Eqs. (3) and (4):

$$C_e\frac{\partial T_e}{\partial t} = \nabla\cdot(-k_e\nabla T_e) - G(T_e - T_l) + Q \tag{3}$$

$$C_l\frac{\partial T_l}{\partial t} = \nabla\cdot(-k_l\nabla T_l) + G(T_e - T_l) \tag{4}$$

where the subscript e refers to electrons and subscript l refers to the lattice; C is the specific heat; k is the thermal conductivity, G is the coupling coefficient of the energy exchange between electron and lattice subsystems.

The electron specific heat C_e [44] was calculated by Eq. (5):

$$C_e = C_{e0}T_{e0} = \frac{\pi^2 N k_B T_e}{2T_F} \tag{5}$$

where C_{e0} is the electron specific heat constant; N is the density of the free electron; k_B is the Boltzmann constant; T_F is the Fermi temperature.

The electron thermal conductivity k_e [45] was calculated by Eq. (6):

$$k_e = \frac{C_e v_F^2 \tau}{3} \tag{6}$$

where v_F is the Fermi velocity, τ is the electron momentum relaxation time.

The thermal physical parameters of the lattice system were taken from that of the copper.

3. Result and discussion

3.1. Effect of laser process parameters

The laser absorptivity of the copper foil at various wavelengths was shown in Fig. 3 (a). The surface morphology of the copper foil and its laser absorptivity were affected by varying pulse numbers. The absorptivity of the copper foil under different laser pulse numbers at 520 nm wavelength was shown in Fig. 3 (b).

The value of R in Equation (2) was set according to the measurement results.

Fig. 4 (a) illustrated the effect of laser power on shear strength. The scanning speed and focal position were maintained at 4 mm/s and −300 μm, respectively. At laser power lower than 0.5 W, the molten materials were insufficient to completely fill the gap and form an effective connection. The maximum shear strength of 17.19 MPa was obtained at 2.0 W. The lattice temperature exceeded 10000 K (Fig. 3d). Both excessive and insufficient laser power resulted in decreased shear strength. Higher than 3.0 W, the femtosecond laser energy was nonlinearly absorbed as it propagated through the glass, leading to modification and ablation within the glass. The laser energy and focal point location were deviated away from the optimal parameters, decreasing the shear strength dramatically. The detailed discussion was in Section 3.2.

Fig. 4 (b) illustrated the effect of scanning speed on shear strength. The laser power and focal position were maintained at 2.0 W and −200 μm, respectively. Welding strength was highly dependent on changes in scanning speed. The maximum shear strength of 13.66 MPa was obtained at 4 mm/s. At speed faster than 10 mm/s, the shear strength remained relatively constant at approximately 5 MPa.

Fig. 4 (c) illustrated the effect of focal position on shear strength. The scanning speed and laser power were maintained at 2 mm/s and 2 W, respectively. The positional reference was defined as positive towards the glass and negative towards the copper direction. The formation of welds was unsuccessful when the laser was focused above the interface (focal position > 0 μm), where the laser with high power density was absorbed nonlinearly by the glass. An ablation area in the glass formed, preventing the laser beam from propagating. Welding was effective when the focal point was positioned within the copper (focal position < 0 μm). The maximum shear strength of 11.8 MPa was obtained at −300 μm, as shown in Fig. 4 (c). When the focal position was below −300 μm, the laser power density incident on the metal surface decreased, leading to a reduction in welding intensity.

3.2. Welds morphology

The welded sample was separated by the tensile testing machine. Welds on the surfaces of the copper foil and glass were observed by SEM and EDS, as shown in Fig. 5. The welds depicted in each figure (labeled as #1, #2, #3, #4, and #5) were formed utilizing various laser welding parameters.

Fig. 5 (a) and (g) showed the welds on copper foil and glass surfaces under the laser power of 1, 2, 3, 4, and 5 W, respectively. Upon reaching the copper foil surface, the laser energy was absorbed by free electrons via the inverse Bremsstrahlung process. Rapid energy relaxation and thermalization occurred within the electronic subsystem on the femtosecond timescale [46,47]. Hot electrons diffused locally. The electron temperature increased almost instantaneously, reaching tens of thousands of Kelvins (Fig. 3c). This resulted in a transient non-thermal equilibrium between electron and the lattice within the materials. Finally, energy was transferred from hot electrons to the lattice, owing to the photoelectron coupling, as shown in Fig. 3 (d). The material was ablated by femtosecond laser pulses with a high repetition frequency. Due to the narrow gap between the copper foil and silica glass, the rapidly generated plasma and vapor expanded but were confined. The redeposited material interacted with the adjacent substrate, resulting in the formation of a stable connection. Increased laser power led to

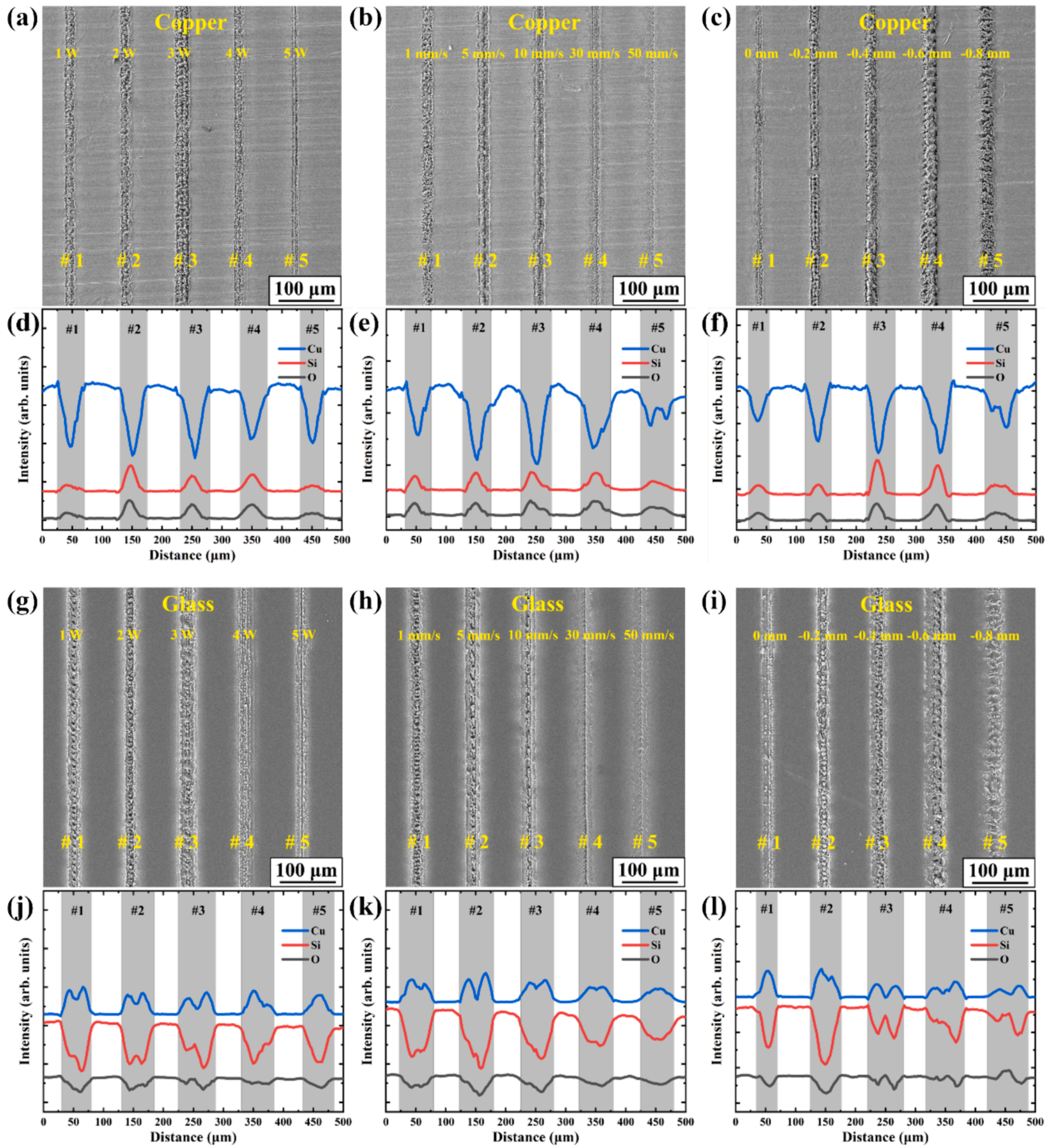

Fig. 5. Morphology of welds under different laser parameters on the copper foil surface: (a) laser power, (b) scanning speed, and (c) focal position; on glass surface: (g) laser power, (h) scanning speed, and (i) focal position; element distribution on the surface by EDS: (d), (e), (f), (j), (k), and (l) was the result of (a), (b), (c), (g), (h), and (i), respectively.

greater material ablation. The width of the welds and the residual mixture of copper and glass increased after separation. However, for laser powers exceeding 3.0 W, the width of the welds decreased. Strong reflection occurred due to excessive laser energy reaching the copper foil surface. Modification and ablation regions resulting from reflection were observed within the glass (Fig. 6).

Fig. 5 (b) and (h) showed the welds on copper foil and glass surfaces at the scanning speed of 1, 5, 10, 30, and 50 mm/s, respectively. The welds width decreased with increasing scanning speed (5~10 mm/s). Excessive scanning speed reduced the power density received by the material. Less material was ablated, and the remaining mixture on the substrate was reduced. However, at the scanning speed of 1 mm/s, modification and ablation regions caused by excessive laser energy concentration were observed near the upper surface, as shown in Figure S2. Subsequent laser pulses were scattered by the modification region, leading to a decrease in energy absorption by the copper surface.

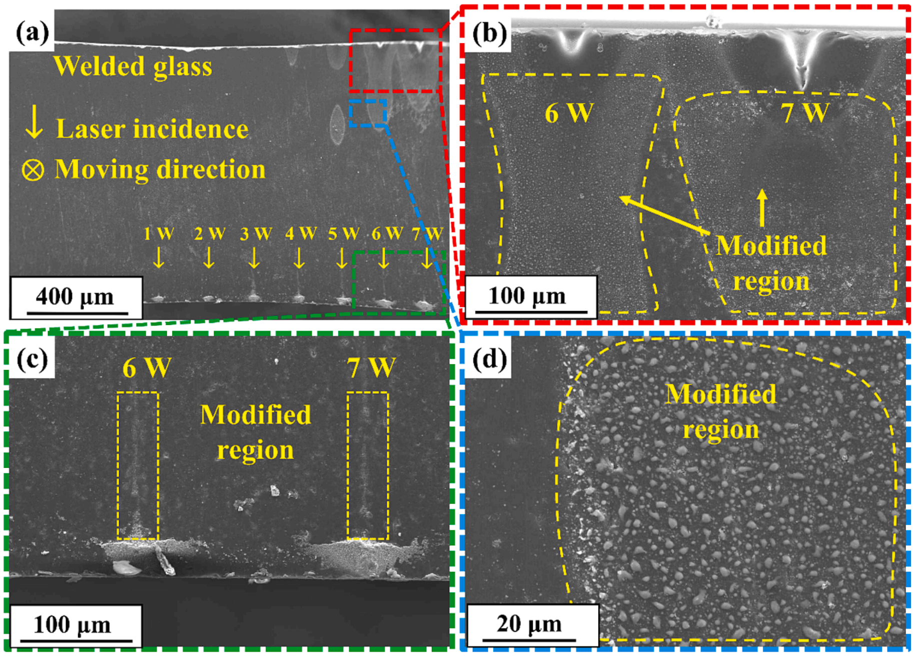

Fig. 6. (a) Cross section of the welded glass under different laser power (1 ~ 7 W); modification and ablation region (b) near the upper surface (red rectangle), (c) near the lower surface (green rectangle), and (d) in the middle (blue rectangle).

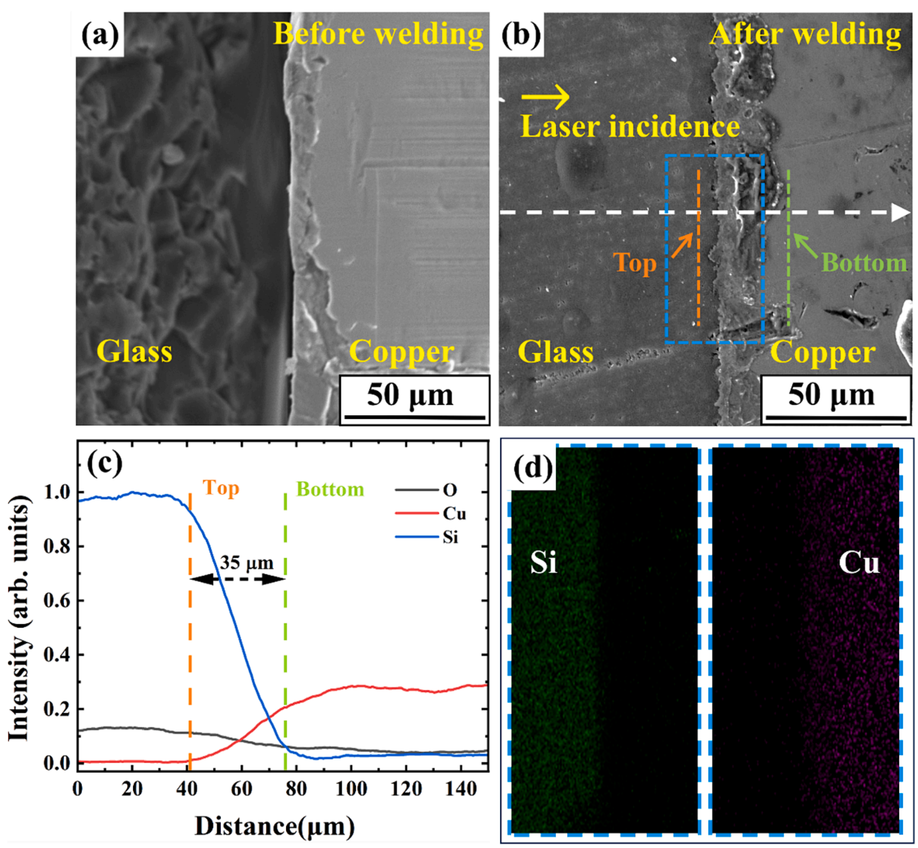

Fig. 7. Cross section of the welds: (a) before welding, (b) after welding; elemental diffusion of the welds by EDS: (c) line scan, (d) map scan.

The weld width was narrower, and element mixing was poorer at 1 mm/s compared to 5 mm/s, consistent with the description in Fig. 5(b).

Fig. 5 (c) and (i) showed the welds on copper foil and glass surfaces at focal position of 0, −0.2, −0.4, −0.6, and −0.8 μm, respectively. At the focal position of 0 μm, a narrower weld width and reduced residual mixture resulted in lower welding strength. The weld width increased as the focal position was shifted downward. However, power density was reduced by largening laser spot. Wider welds with minimal residual mixture contributed to the rapid decrease in welding strength (below −0.6 μm).

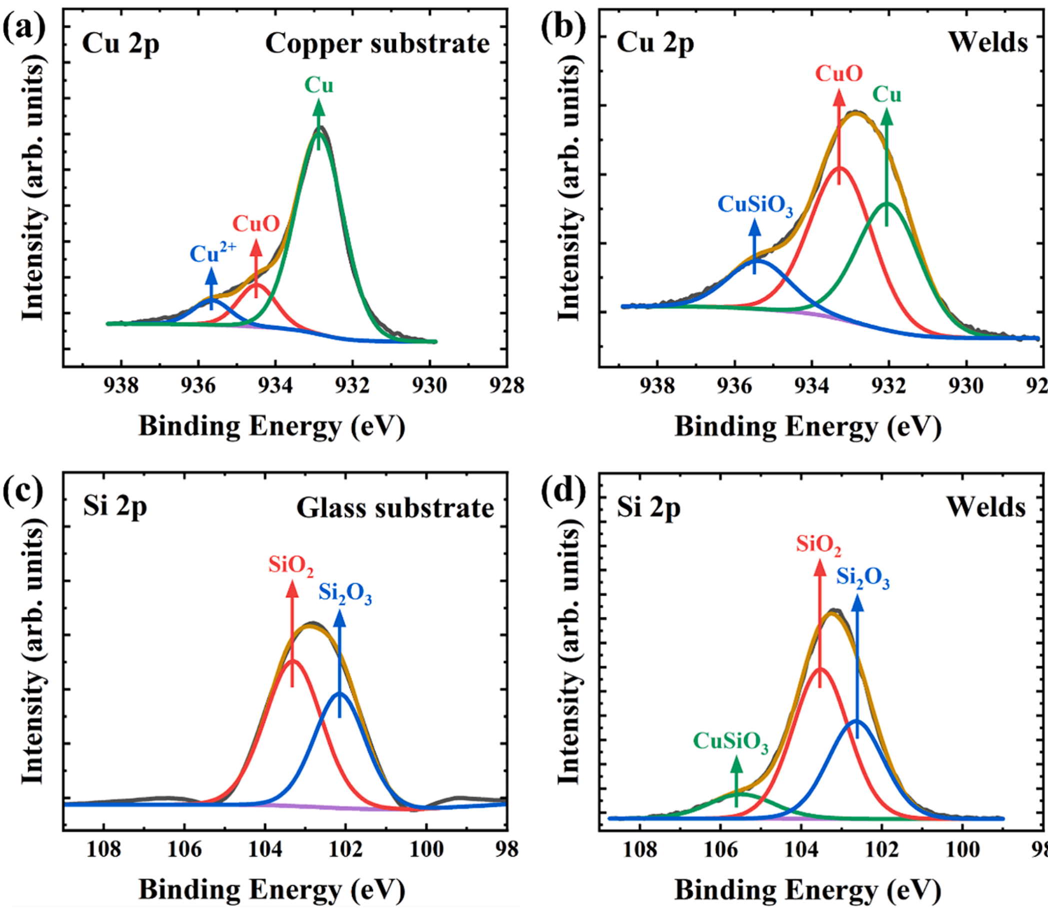

Fig. 8. High-resolution XPS spectra of Cu of the (a) copper substrate, (b) welds; spectra of Si of the (c) glass substrate, (d) welds.

The cross-sections of the welded glass under various laser powers (1 ~ 7 W) was shown in Fig. 6. The unfocused femtosecond laser exceeded the glass breakdown threshold, leading to an ablation area on the upper surface, as shown in Fig. 6 (b). Copper is a highly reflective material. At the laser powers exceeding 3.0 W, a modification region was formed by femtosecond laser pulses (reflected by copper into the glass), as shown in Fig. 6 (c) and (d). Subsequent laser pulses were scattered by the modification region, leading to decreased welding quality, consistent with the observations in Figs. 4 (a) and 5 (a). Two control groups were established to validate the effect of copper reflection: (1) Silica glass was positioned at the same focal point without copper foil, as shown in Figure S3. (2) Stainless steel with a rough surface (reflectivity < 40 %) was placed beneath the silica glass, as shown in Figure S4. Materials were processed using the same laser parameters, and no modification regions were observed after laser irradiation in either condition.

The cross-section of the welds and elemental distribution were observed by using SEM/EDS, as shown in Fig. 7. The surface of the copper foil was relatively rough, as shown in Fig. 7 (a). A connection between the copper foil and silica glass was established after laser welding. The welding strength was enhanced by the mechanical interlocking structure observed at the interface, as shown in Fig. 7 (b).

EDS was used to characterize the elemental distribution of the cross-section by scanning along the white arrows in Fig. 7(b), with the results depicted in Fig. 7(c). The depth of elemental diffusion was 35 μm. The mechanical interlocking structure and elemental diffusion are key factors contributing to welding stability.

3.3. Microstructure and chemical reaction

The microstructure and the chemical reactions during the welding process were characterized by XPS, shown in Fig. 8. The standard binding energy (BE) of 284.8 eV for carbon C 1 s was used to correct all spectral peaks, thereby eliminating potential perturbations (charge effects, surface state, etc.). The core levels of Cu 2p and Si 2p in the welds and substrates were analyzed in Fig. 8, while BE of Cu and Si in the samples were shown in Table S5.

The Cu 2p spectra of the copper substrate was deconvoluted into three peaks (932.9, 934.5, and 935.7 eV), corresponding to the Cu, CuO, and Cu^{2+} in copper oxide, respectively, as shown in Fig. 8 (a). After welding, three distinct Cu species were observed in the welds. Peaks at 932.0, 933.2, and 935.3 eV were attributed to Cu, CuO, and $CuSiO_3$ [48–50], respectively, as shown in Fig. 8 (b). The interaction with the femtosecond laser resulted in the breaking of metallic and Si-O bonds. Cu–O-Si bonds formed as a result of mixing redeposited glass and copper.

The peaks of the silica glass substrate were attributed to SiO_2 and Si_2O_3 [51], respectively, as shown in Fig. 8 (c). The Si 2p spectra of the welds was deconvoluted into three peaks (102.6, 103.5, and 105.5 eV), attributed to the Si_2O_3, SiO_2, and $CuSiO_3$, respectively, as shown in Fig. 8 (d). Laser energy absorption resulted in the breaking of Si-O bonds in the silica glass. Active oxygen atoms with unsaturated coordination were generated. These atoms subsequently bonded with Cu^{2+} ions to form Cu–O-Si bonds. The connection between the two materials was stabilized through the formation of chemical bonds during the welding process.

3.4. Thermal cycling testing

The differing thermal expansions of the two materials induce significant stress during laser welding. Welding stability across different

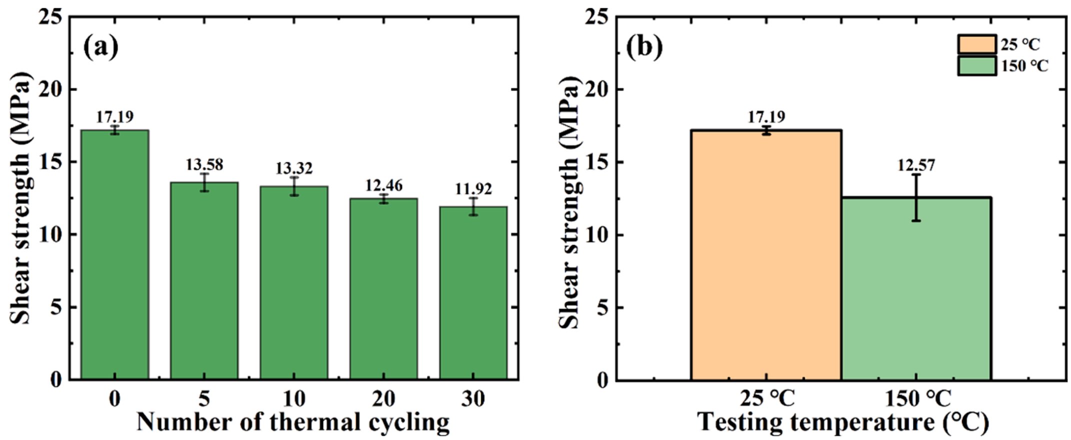

Fig. 9. The shear strength of the welded sample after (a) thermal cycling test (0 – 30 times) between 0 and 100 °C, (b) heating at 150 °C for 3 h.

temperatures was also a critical criterion. Fig. 9 showed the testing of shear strength of the welded samples after thermal cycling and heating.

Welded samples were immersed in water at 0 °C for 1 min, followed by immersion in boiling water at 100 °C for 1 min. The above cycling test was repeated multiple times (0, 5, 10, 20, and 30 cycles). Each experimental parameter was repeated ten times to minimize errors, as shown in Fig. 9 (a). After five thermal cycles, the shear strength decreased to 13.58 MPa. Additional thermal cycles did not significantly reduce the shear strength, which stabilized at approximately 12 MPa. Meanwhile, samples were heated to 150 °C at a rate of 10 °C/min, held for 3 h, and then cooled to room temperature. The shear strength was maintained at 12.57 MPa, as shown in Fig. 9 (b). The connection between two substrates remained stable and effective. The high stability and reliability of our welding method across a broad temperature range were demonstrated.

4. Conclusion

In this work, silica glass and rough copper foil were welded using a femtosecond laser (520 nm). The absorptivity of the copper material was further enhanced by the nonlinear effects of femtosecond pulses and green light from second harmonic generation. No optical contact limitations were required in this welding preprocessing. The morphology of the welds was observed using SEM and CLSM. The microscopic mechanisms, elemental diffusion, and chemical reactions were investigated. Electron and lattice temperatures were simulated using the two-temperature model. The laser energy was absorbed nonlinearly by free electrons and subsequently transferred to the lattice. Material ablation and plasma formation ensued. The redeposited plasma and vapor interacted with copper and glass, leading in the formation of a stable connection. Using the optimized parameters of 2 W laser power, 4 mm/s scanning speed, and −300 μm focal position, a maximum shear strength of 17.19 MPa was achieved. Excessive laser energy, reflected by copper into the glass, led to the formation of a modification region. Subsequent laser pulses were scattered, resulting in a decrease in welding quality. The depth of elemental diffusion measured on the cross-section of the welds was 35 μm. Cu–O-Si bonds were detected on the welds. Welding stability across various temperatures was also assessed. The shear strength remained approximately 12 MPa after thermal cycling from 0 to 100 °C and heating to 150 °C. This study presents an efficient method for laser welding silica glass and rough copper foil, enhancing joint reliability across various industrial applications.

Author contributions

Jingyu Huo and Zirong Zeng made the same contribution: Conceptualization, Data curation, Formal analysis, Software, Writing – original draft, Investigation. **Jiaming Li:** Methodology, Funding acquisition, Supervision, Project administration, Writing – review& editing, Supervision, Project administration. **Huan Yang:** Methodology, Funding acquisition, Supervision, Writing – review & editing. **Qingmao Zhang:** Supervision, Project administration. **Jinhui Yuan, Minghuo Luo, Nan Zhao, and Aiping Luo:** Investigation, Validation.

CRediT authorship contribution statement

Jingyu Huo: Writing – original draft, Software, Formal analysis, Data curation, Conceptualization, Investigation. **Zirong Zeng:** Writing – original draft, Software, Formal analysis, Data curation, Conceptualization, Investigation. **Jinhui Yuan:** Validation, Investigation. **Minghuo Luo:** Validation, Investigation. **Aiping Luo:** Validation, Investigation. **Jiaming Li:** Writing – review & editing, Methodology, Funding acquisition, Supervision, Project administration. **Huan Yang:** Writing – review & editing. **Nan Zhao:** Validation, Investigation. **Qingmao Zhang:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (62475075, 62105105, 12302210); Key Area Research and Development Program of Guangdong Province (2020B090922006); Guangdong Basic and Applied Basic Research Foundation (2021A1515011932, 2023A1515011641, 2023A1111120030); Young Talent Support Project of Guangzhou Association for Science and Technology (QT-2023-007); The Open Project of Guangdong Provincial Key Laboratory of Intelligent Port Security Inspection (2023B1212010011); Guangdong Provincial Engineering Technology Research Center for Materials for Advanced MEMS Sensor Chip (Award No. 2022GCZX005).

References

[1] W.-H. Tuan, S.-K. Lee, Eutectic bonding of copper to ceramics for thermal dissipation applications–A review, J. Eur. Ceram. Soc. 34 (2014) 4117–4130.

[2] T. Mittler, T. Greß, M. Feistle, M. Krinninger, U. Hofmann, J. Riedle, R. Golle, W. Volk, Fabrication and processing of metallurgically bonded copper bimetal sheets, J. Mater. Process. Technol. 263 (2019) 33–41.
[3] T. Chang, M. Sepati, G. Herting, C. Leygraf, G.K. Rajarao, K. Butina, A. Richter-Dahlfors, E. Blomberg, I Odnevall Wallinder, A novel methodology to study antimicrobial properties of high-touch surfaces used for indoor hygiene applications—A study on Cu metal, PLoS One 16 (2021) e0247081.
[4] S. Matsuyoshi, Y. Mizuguchi, A. Muratsugu, H. Yamada, T. Tamaki, W. Watanabe, Welding of Glass and Copper with a Rough Surface using Femtosecond Fiber Laser Pulses, J. Laser Micro/nanoeng. 13 (2018) 21–25.
[5] A. Utsumi, T. Ooie, T. Yano, M. Katsumura, Direct Bonding of Glass and Metal Using Short Pulsed Laser, J. Laser Micro/nanoeng. 2 (2007) 133–136.
[6] F. Marchione, G. Chiappini, P. Munafò, Effect of temperature and relative humidity on the shear performance of double-lap adhesive joints between steel and glass adherends, J. Build. Eng. 45 (2022) 103546.
[7] F. Lambiase, M. Durante, A.D. Ilio, Fast joining of aluminum sheets with Glass Fiber Reinforced Polymer (GFRP) by mechanical clinching, J. Mater. Process. Technol. 236 (2016) 241–251.
[8] L.F. Hu, Y.Z. Xue, H. Wang, Glass-Cu joining by anodic bonding and soldering with eutectic Sn-9Zn solder, J. Alloys Compd. 789 (2019) 558–566.
[9] Z. Yang, Y.Z. Guo, J.W. Stevenson, K.D. Meinhardt, Chemical interactions of barium–calcium–aluminosilicate-based sealing glasses with oxidation resistant alloys, Solid State Ion. 160 (2003) 213–225.
[10] Y. Ding, L. Liu, C. Wang, C. Li, N. Lin, S. Niu, Z. Han, J.A. Duan, Bioinspired near-full transmittance MgF2 window for infrared detection in extremely complex environments, ACS Appl. Mater. Interfaces 15 (2023) 30985–30997.
[11] X. Jia, Y. Chen, L. Liu, C. Wang, Combined pulse laser: Reliable tool for high-quality, high-efficiency material processing, Opt. Laser Technol. 153 (2022) 108209.
[12] X. Jia, K. Li, Z. Li, C. Wang, J. Chen, S. Cui, Multi-scan picosecond laser welding of non-optical contact soda lime glass, Opt. Laser Technol. 161 (2023) 109164.
[13] X. Jia, J. Dong, Y. Chen, H. Wang, G. Zhu, A. Kozlov, X. Zhu, Nanosecond-millisecond combined pulse laser drilling of alumina ceramic, Opt. Lett. 45 (2020) 1691–1694.
[14] G.D. Zhang, J. Bai, W. Zhao, K.M. Zhou, G.H. Cheng, Interface modification based ultrashort laser microwelding between SiC and fused silica, Opt Express 25 (2017) 1702–1709.
[15] H. Chen, J. Duan, Z. Yang, W. Xiong, L.M. Deng, Picosecond laser seal welding of glasses with a large gap, Opt Express 27 (2019) 30297–30307.
[16] J. Huo, B. Zhang, C. Li, M. Luo, T. Chen, L. Guo, A. Luo, N. Zhao, J. Li, Q. Zhang, The mechanism of the welding between silica glass and 304 stainless steel using nanosecond fibre laser, Science and Technology of Welding and Joining 28 (2023) 407–414.
[17] E.H. Penilla, L.F. Devia-Cruz, A.T. Wieg, P. Martinez-Torres, N. Cuando-Espitia, P. Sellappan, Y. Kodera, G. Aguilar, J.E. Garay, Ultrafast laser welding of ceramics, Science 365 (2019) 803–808.
[18] M. Chambonneau, Q. Li, V.Y. Fedorov, M. Blothe, K. Schaarschmidt, M. Lorenz, S. Tzortzakis, S. Nolte, Taming ultrafast laser filaments for optimized semiconductor–metal welding, Laser Photonics Rev. 15 (2021) 2000433.
[19] X.D. Yu, Q. Long, Y. Chen, Y.Y. Liu, C. Yang, Q.B. Jia, C.Y. Wang, Laser transmission welding of dissimilar transparent thermoplastics using different metal particle absorbents, Opt. Laser Technol. 150 (2022) 108005.
[20] P. Sopeña, A. Wang, A. Mouskeftaras, D. Grojo, Transmission laser welding of similar and dissimilar semiconductor materials, Laser Photonics Rev. 16 (2022) 2200208.
[21] M. Zhang, Y.F. Chan, C.J. Chen, Z.L. Qiu, A new sealing technology for ultra-thin glass to aluminum alloy by laser transmission welding method, Int. J. Adv. Manuf. Technol. 115 (2021) 2017–2035.
[22] C.K. Li, J.T. Tan, M.H. Luo, W.J. Chen, Y.X. Huang, J.L. Gu, N. Zhao, J.M. Li, H. Yang, Q.M. Zhang, High shear strength welding of soda lime glass to stainless steel using an infrared nanosecond fiber laser assisted by surface tension, Opt. Lasers Eng. 161 (2023) 107329.
[23] J. Huo, J. Yuan, Q. Chen, M. Luo, J. Lu, J. Xu, C. Li, A. Luo, J. Li, Q. Zhang, Welding reinforcement between silica glass and stainless steel using nanosecond fiber laser with chromium interlayer, Opt. Lasers Eng. 172 (2024) 107877.
[24] L. Mei, Z. Wang, D. Yan, S. Chen, D. Xie, Effect of activating flux on laser penetration welding performance of galvanized steel, Int. J. Adv. Manuf. Technol. 91 (2017) 1069–1078.
[25] I. Miyamoto, K. Cvecek, M. Schmidt, Advances of Laser Welding Technology of Glass-Science and Technology, J. Laser Micro/nanoeng. 15 (2020) 63–67.
[26] J. Zhang, S. Chen, H. Lu, M. Huang, J. Li, L. Guo, Q. Lue, Q. Zhang, The effect of gap on the quality of glass-to-glass welding using a picosecond laser, Opt. Lasers Eng. 134 (2020) 106248.
[27] R.M. Carter, J.Y. Chen, J.D. Shephard, R.R. Thomson, D.P. Hand, Picosecond laser welding of similar and dissimilar materials, Appl Opt 53 (2014) 4233–4238.
[28] Y.P. Wang, Y. Li, S.S. Ao, Z. Luo, D. Zhang, Welding of 304 stainless steel and glass using high-repetition-frequency femtosecond laser, Mater. Res. Express 8 (2021) 106523.
[29] L. Zhang, H. Wu, J.C. Wen, M. Li, X.Y. Shao, X.Q. Ma, Glass to aluminum joining by forming a mechanical pin structure using femtosecond laser, J. Mater. Process. Technol. 302 (2022) 117504.
[30] Y. Ozeki, T. Inoue, T. Tamaki, H. Yamaguchi, S. Onda, W. Watanabe, T. Sano, S. Nishiuchi, A. Hirose, K. Itoh, Direct welding between copper and glass substrates with femtosecond laser pulses, Appl. Phys. Express 1 (2008) 082601.
[31] G.D. Zhang, G.H. Cheng, Direct welding of glass and metal by 1 kHz femtosecond laser pulses, Appl. Opt. 54 (2015) 8957–8961.
[32] Q Lia, G Matthäusa, S Noltea, Glass to copper direct welding with a rough surface by femtosecond laser pulse bursts, Lasers in Manufacturing Conference, (2021).
[33] M. Haubold, A. Ganser, T. Eder, M.F. Zäh, Laser welding of copper using a high power disc laser at green wavelength, Procedia CIRP 74 (2018) 446–449.
[34] E.-M. Dold, E. Kaiser, K. Klausmann, S. Pricking, S. Zaske, R. Brockmann, High-performance welding of copper with green multi-kW continuous-wave disk lasers, High-Power Laser Materials Processing: Applications, Diagnostics, and Systems VIII (2019). SPIE, 179–184.
[35] F. Kaufmann, M. Strugulea, C. Höltgen, S. Roth, M. Schmidt, Seam Properties of Overlap Welding Strategies from Copper to Aluminum Using Green Laser Radiation for Battery Tab Connections in Electric Vehicles, Materials 16 (2023) 1069.
[36] J.J. Chang, B.E. Warner, E.P. Dragon, M.W. Martinez, Precision micromachining with pulsed green lasers, J. Laser Appl. 10 (1998) 285–291.
[37] C. Zhang, N.R. Quick, A. Kar, Microvia drilling with a green laser, International Congress on Applications of Lasers & Electro-Optics, (2007), Laser Institute of America, M1006.
[38] J. Yu, J. Wu, H. Yang, P. Li, J. Liu, M. Wang, J. Pang, C. Li, C. Yang, K. Xu, Extremely sensitive SERS sensors based on a femtosecond laser-fabricated superhydrophobic/-philic microporous platform, ACS Appl. Mater. Interfaces 14 (2022) 43877–43885.
[39] P. Stein, O. García, M. Morales, H.P. Huber, C. Molpeceres, Numerical simulation of laser ablation for photovoltaic materials, Appl. Surf. Sci. 258 (2012) 9288–9291.
[40] J.H. Bunton, J.D. Olson, D.R. Lenz, T.F. Kelly, Advances in pulsed-laser atom probe: Instrument and specimen design for optimum performance, Microsc. Microanal. 13 (2007) 418–427.
[41] A. Suslova, A. Hassanein, Numerical simulation of ballistic electron dynamics and heat transport in metallic targets exposed to ultrashort laser pulse, J. Appl. Phys. 124 (2018) 065108.
[42] I. Miyamoto, K. Cvecek, Y. Okamoto, M. Schmidt, Internal modification of glass by ultrashort laser pulse and its application to microwelding, Appl. Phys. A 114 (2014) 187–208.
[43] E. Majchrzak, B. Mochnacki, Numerical simulation of thermal processes in a domain of thin metal film subjected to an ultrashort laser pulse, Materials 11 (2018) 2116.
[44] L. Jiang, H.-L. Tsai, Improved Two-Temperature Model and Its Application in Ultrashort Laser Heating of Metal Films, J. Heat Transfer 127 (2005) 1167–1173.
[45] R. Fang, D. Zhang, H. Wei, Z. Li, F. Yang, Y. Gao, Improved two-temperature model and its application in femtosecond laser ablation of metal target, Laser Part. Beams 28 (2010) 157–164.
[46] J. Cheng, C.-S. Liu, S. Shang, D. Liu, W. Perrie, G. Dearden, K. Watkins, A review of ultrafast laser materials micromachining, Opt. Laser Technol. 46 (2013) 88–102.
[47] J. Lopez, K. Mishchik, G. Mincuzzi, E. Audouard, E. Mottay, R. Kling, Efficient Metal Processing Using High Average Power Ultrafast Laser, J. Laser Micro/nanoeng. 12 (2017) 296–303.
[48] R. Bird, P. Swift, Energy calibration in electron spectroscopy and the re-determination of some reference electron binding energies, J. Electron. Spectrosc. Relat. Phenom. 21 (1980) 227–240.
[49] S. Bawaked, K. Narasimharao, Structural and catalytic properties of copper silicate nanomaterials, Sci. Rep. 10 (2020) 518.
[50] Q. Li, J. Wu, L. Lv, L. Zheng, Q. Zheng, S. Li, C. Yang, C. Long, S. Chen, Z. Tang, Efficient CO2 electroreduction to multicarbon products at CuSiO3/CuO derived interfaces in ordered pores, Adv. Mater. 2305508 (2023).
[51] D.R. Wheeler, S.V. Pepper, Angle-resolved X-ray photoelectron spectroscopy of epitaxially grown (100) β-SiC to 1300 °C, Surf. Interface Anal. 10 (1987) 153–162.